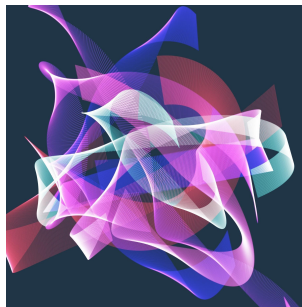
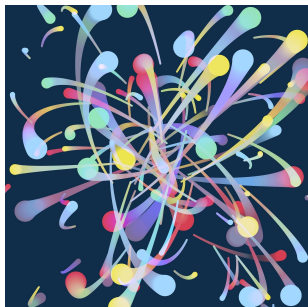
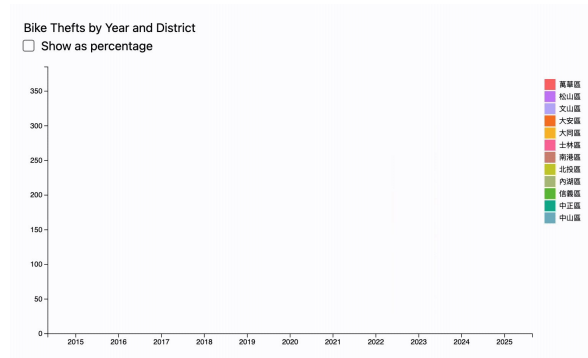
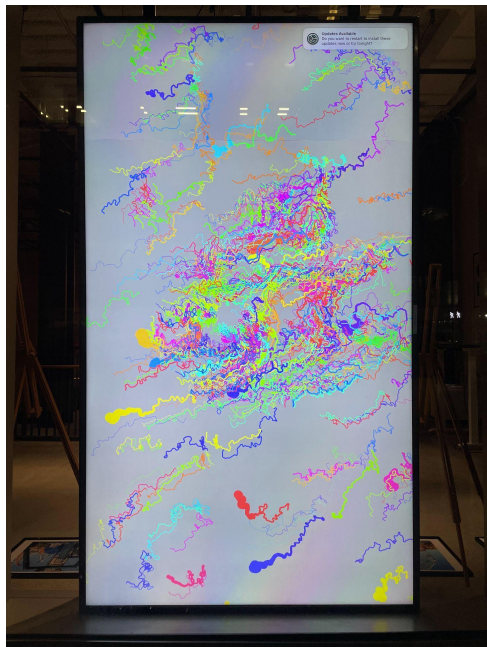


get started with ml5

ccc 2026: by Yu and Chih-Yu



什麼是ml5.js?

ml5.js是一個開源(open-source)、新手友善(beginner-friendly)、在瀏覽器中(browser-based)執行機器學習的JavaScript函式庫

docs.ml5js.org

在網頁上執行機器學習：





+

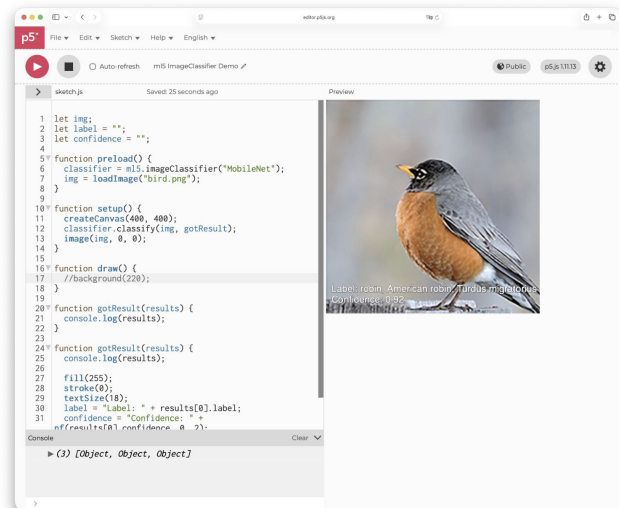


=

Run your sketch

Now, you are ready to see the results! Run your sketch and see if the model can make predictions and provide meaningful outputs. Press the run button on the top left corner of the editor.

You should get something like this:



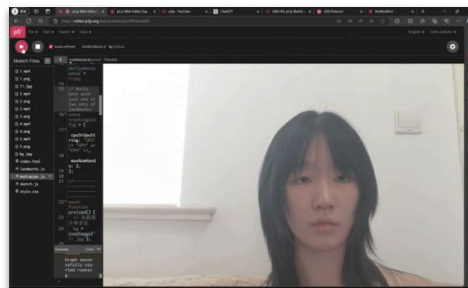
ml5.js能做什麼？



Doodle by Yifan Chen



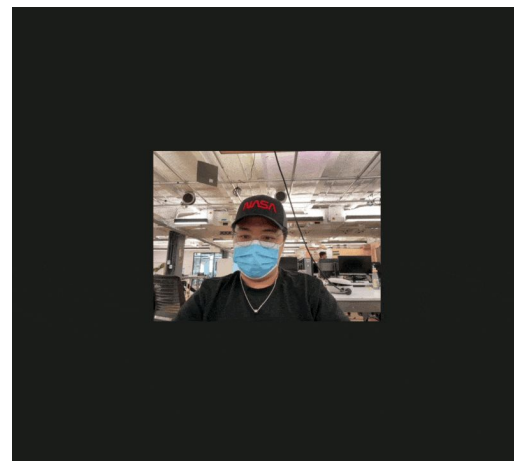
Moving Sign Language by
Matthew Ponon



Bird Bird Bird by Kexin Yang



By ilithya.rocks



Direction Training by Alan Yam



By Jack B. Du

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[DepthEstimation](#)

ml5 + Teachable Machine

[Image + Teachable Machine](#)

Train your own model!

[NeuralNetwork](#)



Getting Started

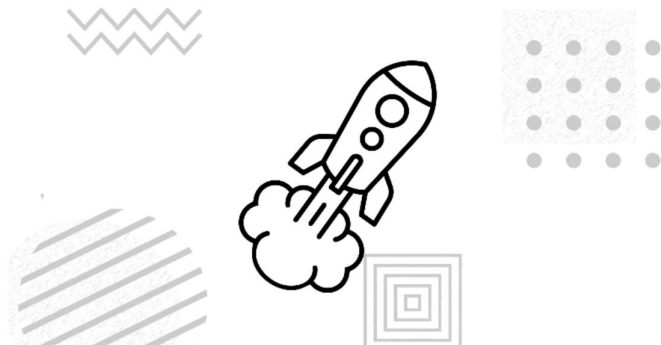


Image Credit: [lki](#) | [Contribute](#) ❤️

Welcome! We're going to walk through how to start using ml5.js by creating a simple image classification program.

This page will cover how to:

1. Load a pre-trained ml5.js image classification model
2. Load an image for the model to identify the object in the image
3. Get the results from the model and display them on the canvas

We will be using a p5.js sketch running on the [p5.js web editor](#). To get started, open up the p5.js web editor and create an empty project. Be sure to sign up or log in to your account so that you are able to upload files! This will be necessary later on as we upload images.

手部偵測 : handPose
臉部偵測 : faceMesh
身體偵測 : bodyPose

如何在p5.js project 裡開始使用ml5.js？

在你的index.html或其他.html中的<head></head>裡面貼入
以下script tag

```
<script src="https://unpkg.com/ml5@1/dist/ml5.min.js"></script>
```

Now, let's switch to the index.html file and copy and paste the following CDN link inside the <head> tag.

```
<script src="https://unpkg.com/ml5@1/dist/ml5.min.js"></script>
```

html



Source: docs.ml5js.org/#/

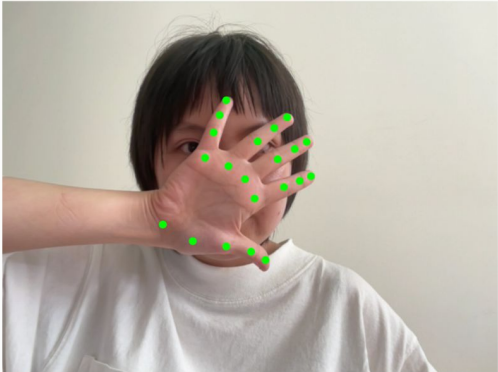
如何使用ml5.js中的模型？(以handPose為例)

p5* File Edit Sketch Help English

Auto-refresh ccc: handPose-keypoints

sketch.js Saved: 25 seconds ago Preview

```
1 let handPose;
2 let video;
3 let hands = [];
4
5 function preload() {
6   // 讀取模型
7   handPose = ml5.handPose({flipped: true});
8 }
9
10 function setup() {
11   createCanvas(640, 480);
12   // 引入video並隱藏
13   video = createCapture(VIDEO, {flipped: true});
14   video.size(640, 480);
15   video.hide();
16   // 開始偵測來自camera video的手部data
17   handPose.detectStart(video, gotHands);
18 }
19
20 function draw() {
21   // 繪製video影像
22   image(video, 0, 0, width, height);
23
24   // 繪製手部keypoints
25   for (let i = 0; i < hands.length; i++) {
26     let hand = hands[i];
27     for (let j = 0; j < hand.keypoints.length; j++) {
28       let keypoint = hand.keypoints[j];
29       fill(0, 255, 0);
30       noStroke();
31       circle(keypoint.x, keypoint.y, 10);
32     }
33   }
34 }
35
36 // Callback 獲取手部的data
37 function gotHands(results) {
38   // save the output to the hands variable
39   hands = results;
40 }
41
```



Console Clear

什麼是callback function (callback函式)?

callback function是指將函式當作參數(parameter)傳入另一個函式, **等某個動作完成後, 再呼叫這個函式**

例如: 載入模型或等 webcam 準備好, 這些事情不會馬上完成
所以用 callback 讓程式**完成後再通知我、完成後再執行**

if/else statements裡面執行的code也可以算是一種callback

什麼是callback function (callback函式)?

```
handPose.detectStart(video, gotHands)
```

```
// Callback 獲取手部的data
```

```
function gotHands(results) {  
  // 把偵測到的data存入hands變數  
  hands = results  
}
```

Code walkthrough: [handpose-handpaint](#)

如何在p5 sketch中擁有不同圖層？

在p5 sketch中，有時候我們希望sketch某些繪製的影響能夠不斷更新、不留痕跡，有些痕跡我們希望留下

這種情況，可以使用`createGraphics()`在sketch中建立新的圖層，並設定這個圖層不用更新、清理(`clear`)

使用createGraphics()新增一個圖層

```
let painting
...

painting = createGraphics(640, 480)
...

painting.circle(x,y,d)
...

image(painting, 0, 0)
```

Code walkthrough: [faceMesh-hearteyes](#)

Code walkthrough: ccc:
faceMesh-hearteyes-tilted
如何讓圖形隨著臉部旋轉？

```
push()  
translate()  
rotate()  
pop()
```

Code walkthrough: [faceMesh-keypoints](#)

Code walkthrough:
faceMesh-hearteyes-tilted
如何讓圖形隨著臉部旋轉？

push() pop(): 讓迴圈中個別的圖形變化(translate, rotate, scale
等)不會互相影響

translate(): 將圖形變化的anchor point從sketch左上角移動至個別
圖形的位置

Code walkthrough: [facemesh-mouthfire](#)

如何讓sketch隨window大小變化並維持圖形比例？

```
createCanvas(windowWidth, windowHeight)

...
let videoW = video.width
let videoH = video.height
let w = width
let h = w * (videoH/videoW)

// Scale factor from video space -> canvas space
let scaleX = w / videoW
let scaleY = h / videoH
image(video, 0, 0, w, h);

...
function windowResized() {
  resizeCanvas(windowWidth, windowHeight);
}
```

Code walkthrough: [facemesh doodle](#)

Code walkthrough: `m15` `body` `highlight`

使用beginShape(), endShape(), vertex()繪製不規則形狀

```
beginShape();  
vertex(pose.left_shoulder.x, pose.left_shoulder.y);  
vertex(pose.right_shoulder.x, pose.right_shoulder.y);  
vertex(pose.right_hip.x, pose.right_hip.y);  
vertex(pose.left_hip.x, pose.left_hip.y);  
endShape(CLOSE);
```


使用drawingContext調整canvas材質

```
drawingContext.shadowBlur = 25  
drawingContext.shadowColor = "rgb(252, 186, 3)"  
  
// 寫成function  
function glow(r, g, b) {  
    drawingContext.shadowBlur = 25;  
    drawingContext.shadowColor = `rgb(${r}, ${g}, ${b})`;  
}
```